

Physics - Gravitation -

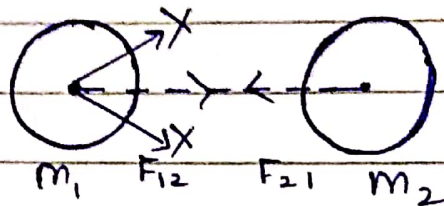
Date

*) Newton's law of gravitation -

$$F \propto m_1 m_2 \quad \text{and} \quad F \propto \frac{1}{r^2}$$

$$\therefore F = \frac{G M_1 M_2}{r^2} \quad \text{where } G = \text{Universal gravitation constant}$$

i.e.,



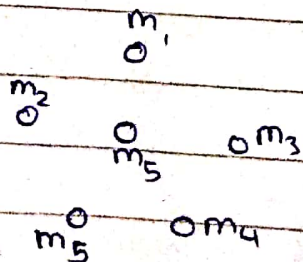
$$\{ G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2 \}$$

- The force is central.
- The force is Attractive
- The force is independent of medium.
- $\vec{F}_{12} = -\vec{F}_{21}$
- Valid Only for point masses.
- Significant only for heavy bodies
- Gravitation is an Internal force.

*) Principle of Superposition :-

If no. of forces are acting on a point mass, then the net force on point is the vector addⁿ of all the forces

i.e. :-



$$\left\{ \begin{aligned} F_{\text{net}}(m_1) &= F_{m_1} + F_{m_1 m_2} + F_{m_1 m_3} \dots \\ &F_{m_1 (n-1)} \end{aligned} \right\}$$

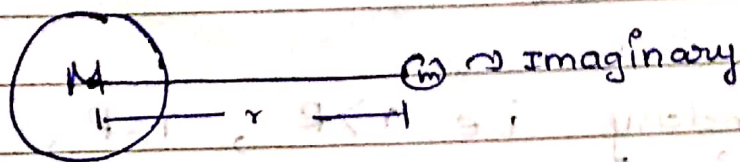
*.) Gravitational Field :-

Gravitational field Strength / Intensity ($\vec{E}$) :-

i.e. $\vec{E} = \frac{\vec{F}}{m}$, unit = N/kg or m/s^2 .

for multiple GFS :- $\{\vec{E}_n = \vec{E}_1 + \vec{E}_2 + \dots\}$

• Gravitational field due to Mass point mass 'M' -



$$\therefore F_{mM} = \frac{GmM}{r^2}$$

$$\therefore \vec{E} = \frac{\vec{F}}{m} = \frac{GmM}{r^2 m} = \left\{ \frac{-GM}{r^2} \right\}$$

• To find force at a point mass :-

$F = mE$ where E is Gravitational field strength.

• General formula to calculate Gravitational field at any angle ϕ for a point mass :-

$$\left\{ E = \frac{GM}{R^2 \phi} \frac{2 \sin \phi}{2} \right\}$$

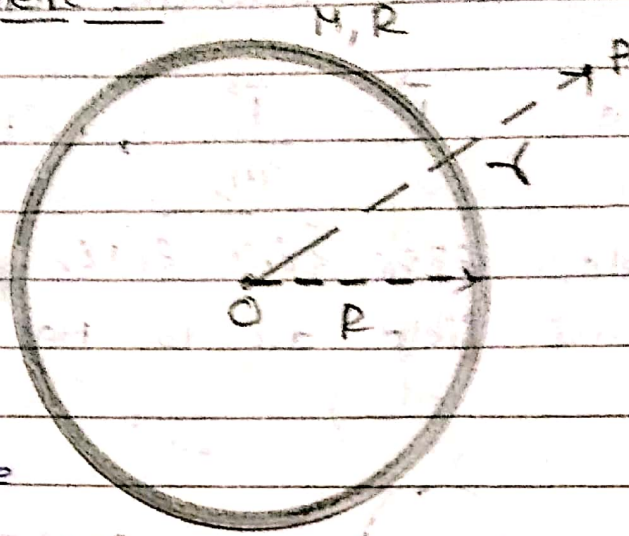
*.) Gravitational field due to hollow + Solid Sphere:-

1.) For hollow Sphere:-

mass = M .

radius = R .

Three points
are chosen



• Outside circle

• Inside circle

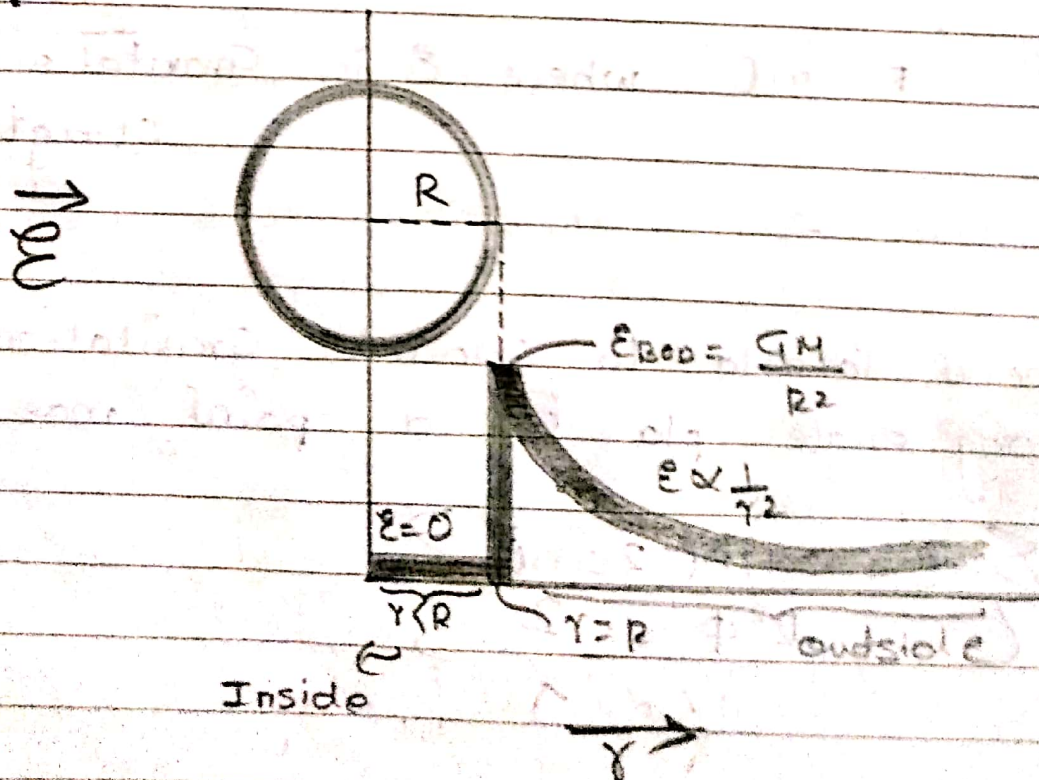
• At boundary i.e. $r > R$, $r = R$, $r < R$

∴ Gravitational field,

• Outside: $E_{out} = \frac{GM}{r^2}$, • $E_{ood} :- E = \frac{GM}{r^2}$

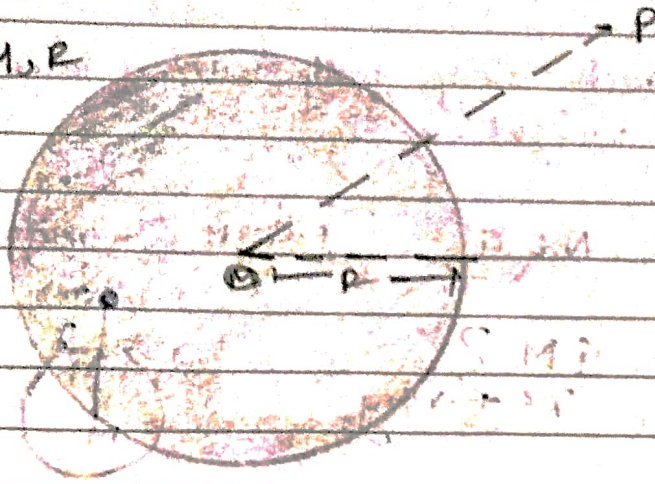
• Inside, i.e. $r < R$, $E = 0$ [Gauss law of gravitation]

Graph:-



2. For solid cylinder -

M, R



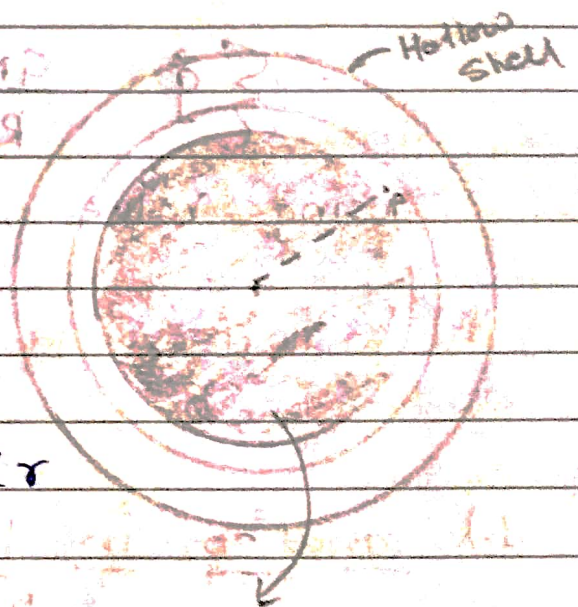
Outside $E = \frac{GM}{r^2}$, $E \propto \frac{1}{r^2}$

At Boundary $E = \frac{GM}{R^2}$ {constant}

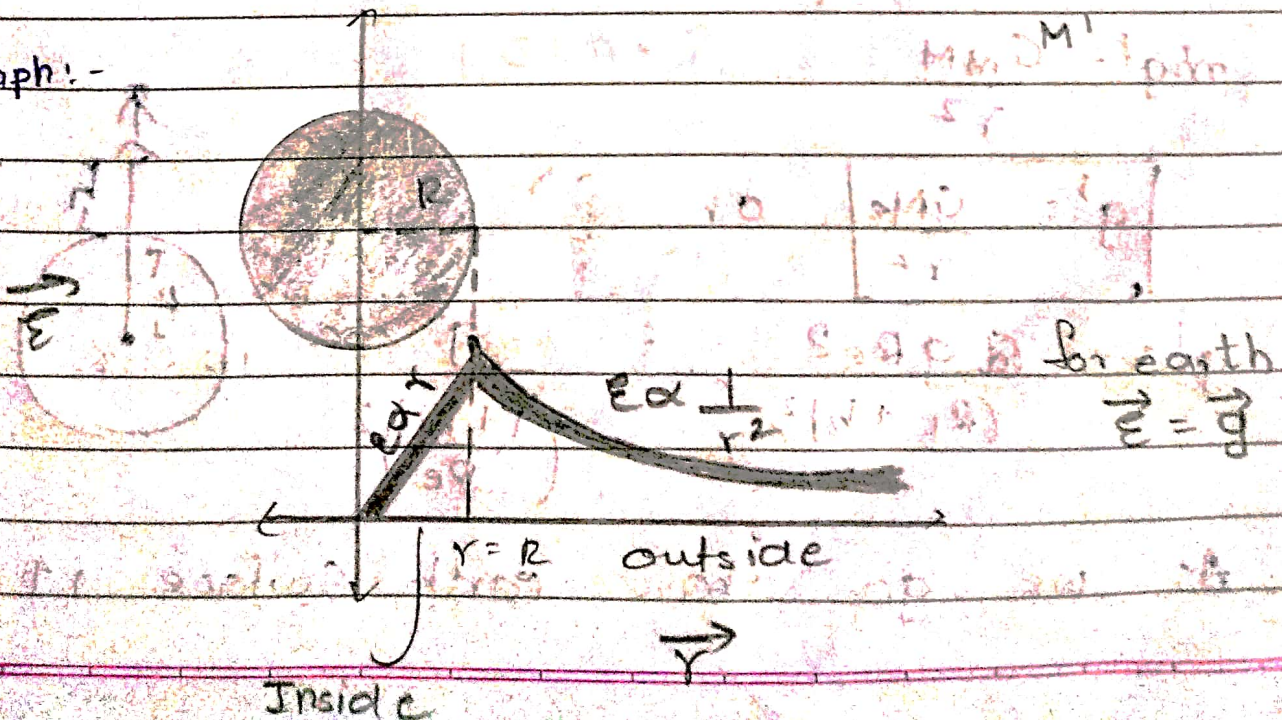
Inside $E_{\text{Hollow}} + E_{M'}$
 $= 0 + E_{M'}$

For M' , Point P is on Boundary

$E_{M'} = \frac{GM'}{r^2} = \frac{4G\rho\pi r}{3}$, $E \propto r$



Graph:-



$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

$$M = 6 \times 10^{24} \text{ kg}$$

$$R = 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$$

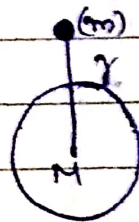
Page No.	
Date	

*) Acceleration due to gravity :- $(\vec{g}) = 9.8 \text{ m/s}^2 \approx 10 \text{ m/s}^2$

Earth attracts every body with a force equal to $= mg$ called weight.

applying N.L.G, $\frac{F = GmM}{r^2} = mg$

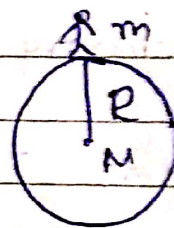
$$\text{vary } \left\{ g = \frac{GM}{r^2} \right\} \text{ vary}$$



The value of $\vec{g} = 9.8 / 10 \text{ m/s}^2$ is taken at surface of earth.

i.e. $r = R$

$$\therefore \left\{ \vec{g} = \frac{GM}{R^2} \right\}$$



using values, we get $\vec{g} = 9.8 \text{ m/s}^2$ or 10 m/s^2

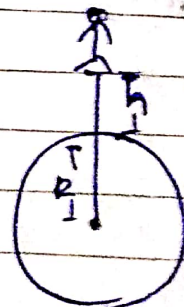
*) Variation in $\vec{g}$:-

1.) Variation in height / depth from surface :- $(\vec{g})$:-

$$mg' = \frac{GmM}{r^2}$$

$$\boxed{g' = \frac{GM_e}{r^2}} \text{ or}$$

$$g' = \frac{g R_e^2}{(R_e + h)^2} = \frac{g}{\left(1 + \frac{h}{R_e}\right)^2}$$



As we go above earth surface, $r \uparrow$, $g' \downarrow$

$$\left\{ g' = \frac{g}{\left(1 + \frac{h}{R_e}\right)^2} \right\}$$

• Special Case:-

If $h \ll R_e$, then,

$$\left\{ g' = g \left(1 + \frac{h}{R_e}\right)^{-2} \approx g \left(1 - \frac{2h}{R_e}\right) \right\} \because (1+x)^{-n} \approx (1-nx)$$

*.) Variation in 'g' as we go Below Earth's Surface :-

$$mg' = \frac{GMm}{r^2}$$

$$\therefore \left\{ g' = \frac{GM}{r^2} \right\}$$

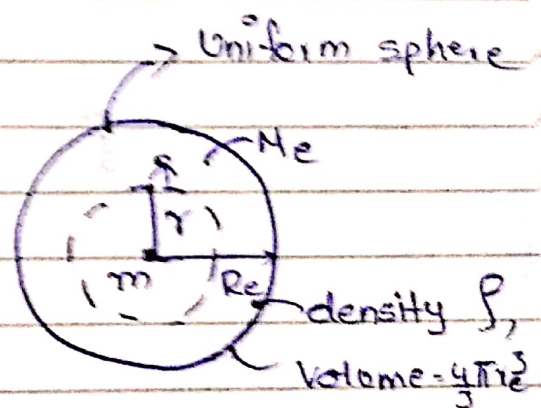
$$\therefore g' = G \times \rho \times \frac{4}{3} \pi r^3 \div r^2$$

$$\therefore \left\{ g' = \frac{4}{3} G \rho \pi r \right\}$$

again

$$g' = \frac{4}{3} G \times \frac{M_e}{\frac{4}{3} \pi R_e^3} \times \frac{4}{3} \pi r$$

$$= \left\{ g' = \frac{GM_e r}{R_e^3} \right\}$$



again,

$$g' = \frac{GM_e r}{R_e^3}$$

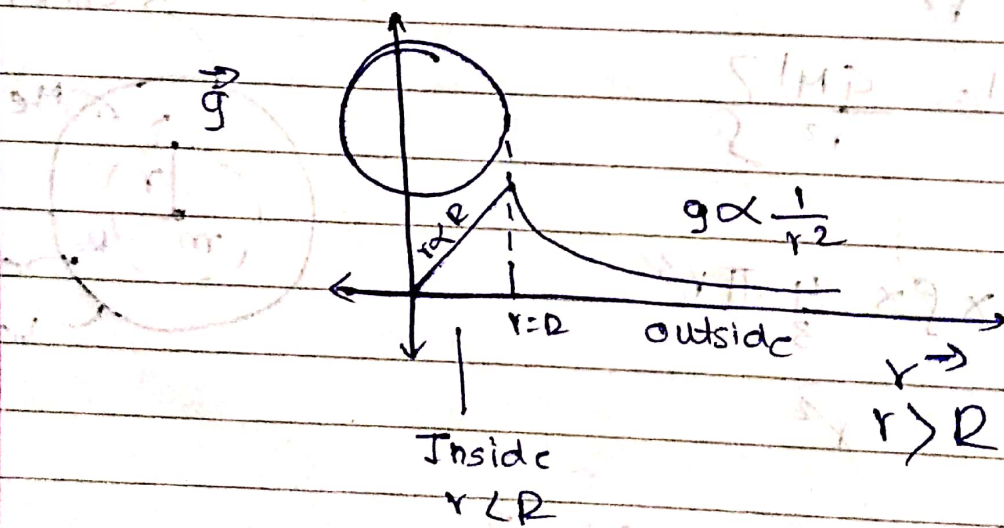
$$g' = \frac{g R_e^2 r}{R_e^3}$$

$$\left[\because g = \frac{GM_e}{R_e^2} \right]$$

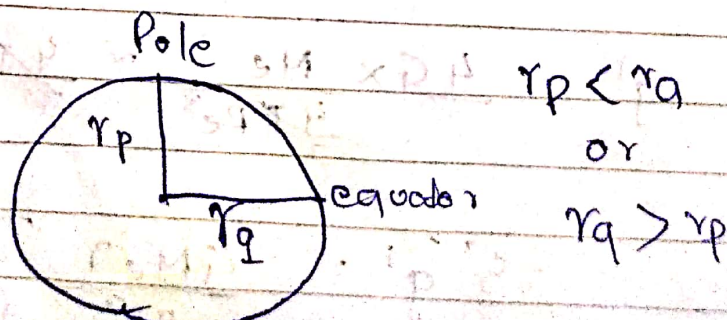
$$\therefore g' = \left(\frac{g}{R_e} \right) r$$

$$\left\{ \because g' = g \left(1 - \frac{h}{R_e} \right) \right\} \because (r = R_e - h)$$

(JEE main) :-



Variation in 'g' due to Shape of Earth

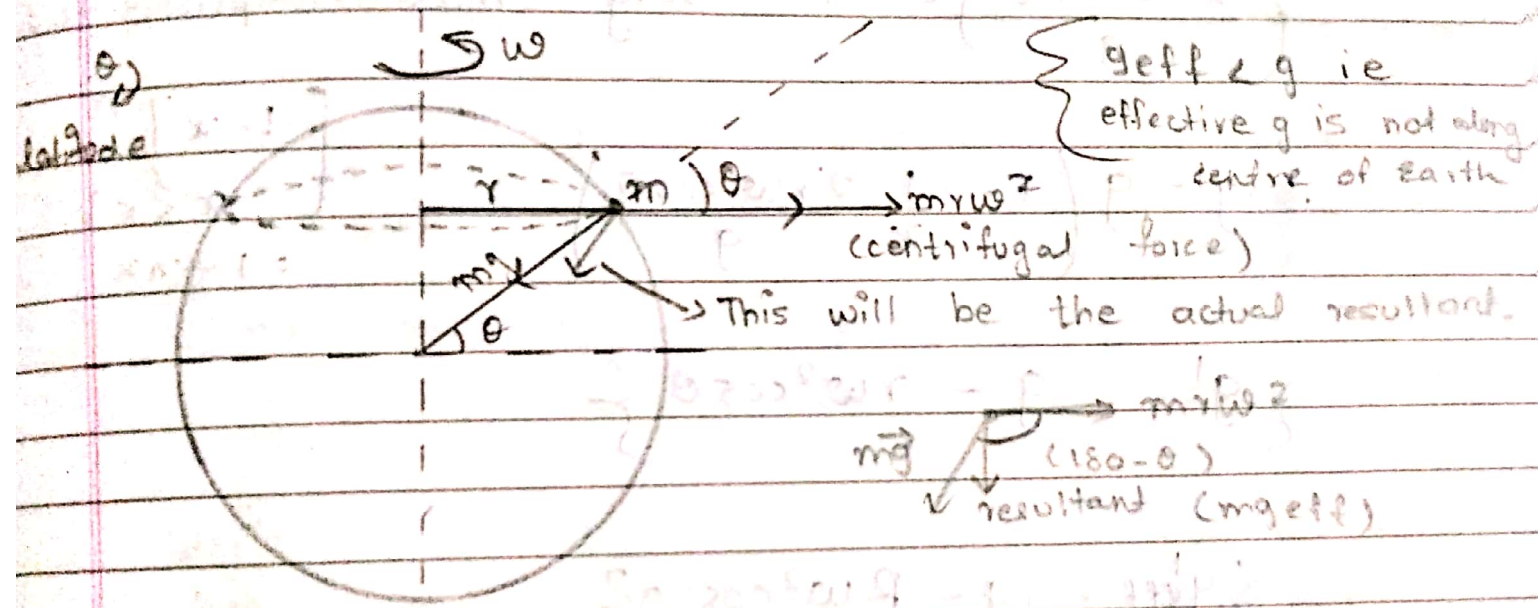


$\{g_{eq} < g_{pole}\}$ (experimentally proved)

The results were true experimentally but did not confirm with calculation.

The problem was with the concept of 'shape' so they introduced a new concept below.

*.) Variation of ' $\vec{g}$ ' due to Rotation of Earth :-



$$\therefore mg_{eff} = mg + mr\omega^2$$

$$\therefore \{ \vec{g}_{eff} = -\vec{g} + \vec{r}\omega^2 \}$$

Using Vector Addⁿ property, resultant, will be

$$\therefore g_{eff} = \sqrt{(g)^2 + (r\omega^2)^2 + 2(g)(r\omega^2)\cos(180-\theta)}$$

$$g_{eff} = \sqrt{g^2 + r\omega^4 - 2gr\omega^2\cos\theta}$$

$\therefore \omega \rightarrow$ very small $\therefore r\omega^4 \rightarrow 0$
it's neglected.

$$\therefore g_{eff} = \sqrt{g^2 - 2gr\omega^2 \cos \theta}$$

$$= g \sqrt{1 - \frac{2r\omega^2 \cos \theta}{g}}$$

$$\therefore g' = g \left(1 - \frac{2r\omega^2 \cos \theta}{g} \right)^{\frac{1}{2}}$$

$\left(\frac{2r\omega^2 \cos \theta}{g} \right)^{\frac{1}{2}}$ is very small compared to 1

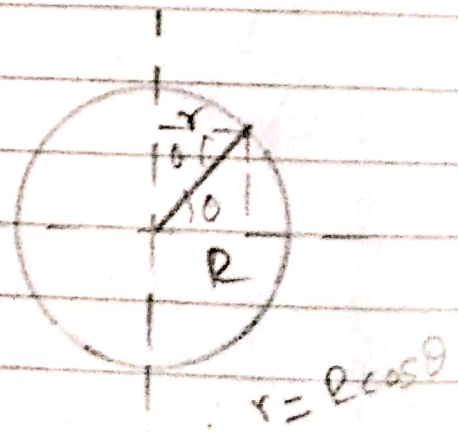
$$\therefore g' = g \left(1 - \frac{1}{2} \frac{2r\omega^2 \cos \theta}{g} \right)$$

$$\left[\begin{array}{l} [1-x]^n \\ x \ll 1 \\ = 1 - nx \end{array} \right]$$

$$\{ g' = g - r\omega^2 \cos \theta \}$$

$$\{ g'_{eff} = g - R\omega^2 \cos^2 \theta \}$$

(at any where)



• At equator, $\theta = 0$

$$\therefore \{ g'_{eff} = g - R\omega^2 \} \rightsquigarrow \text{min.}$$

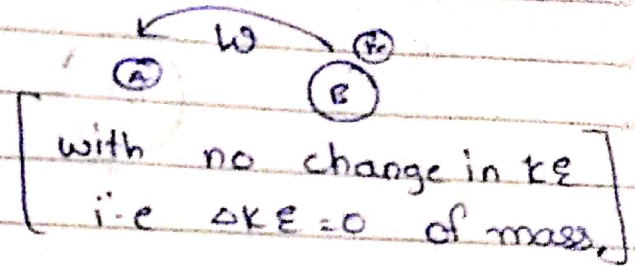
• At pole, $\theta = 90^\circ$

$$\therefore \{ g' = g \} \rightsquigarrow \text{max.}$$

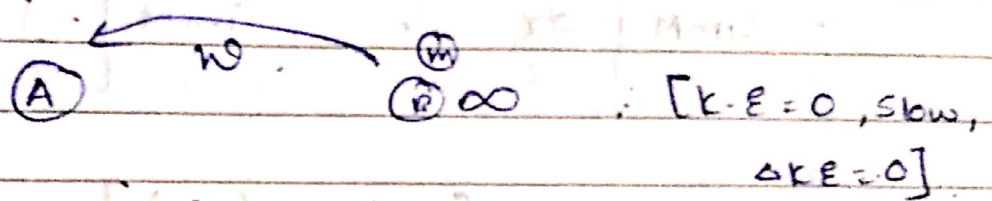
* Gravitational Potential :-

Potential Difference :-

$$\left\{ V_A - V_B = \frac{W_{B \rightarrow A}}{m} \right\}$$



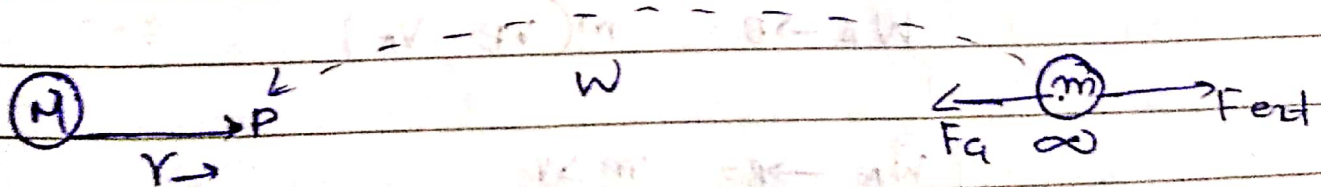
Potential at a point :- [we take a frame of reference at ∞] i.e. $V_{\infty} = 0$.



$$V_A - V_{\infty} = \frac{W_{\infty \rightarrow A}}{m}$$

$$V_A = \frac{W_{\infty \rightarrow A}}{m}$$

* Gravitational potential due to a mass 'M' :-



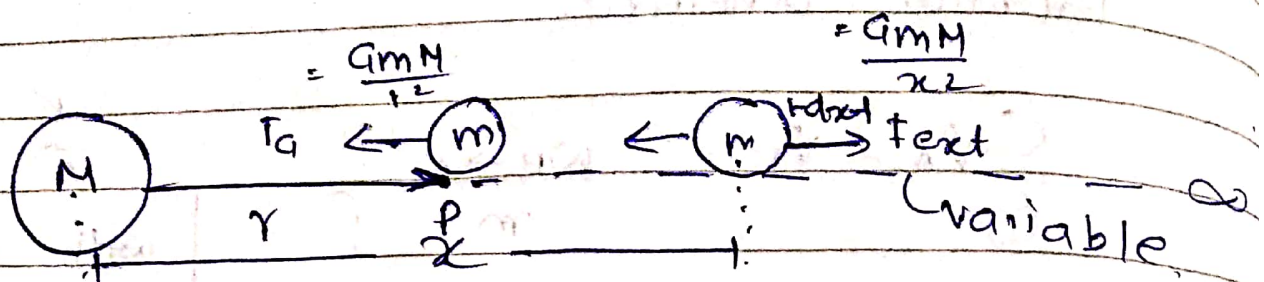
$$V_P = \frac{W_{\infty \rightarrow P}}{m}$$

work done by external agent.

Scalar Quantity

$$\left\{ V_P = -\frac{GM}{r} \right\}$$

Derivation :-



$$W_{ext} = \int_r^\infty F \cdot dx = \int_r^\infty \frac{GmM}{x^2} dx$$

$$= GmM \int_r^\infty \frac{dx}{x^2} = GmM \left[-\frac{1}{x} \right]_r^\infty$$

$$= GmM \left[-\frac{1}{\infty} - \left(-\frac{1}{r} \right) \right] = \left\{ \frac{GmM}{r} \right\} \left[\because \frac{1}{\infty} = 0 \right]$$

$$\therefore \text{G.P. at } P = \frac{W_{A \rightarrow B}}{m} = \frac{GmM}{rm}$$

$$\therefore \text{G.P. at } P = -\frac{GM}{r} \text{ due to Mass } M$$

$$\begin{aligned} W_{A \rightarrow B} &= m(v_B - v_A) \\ W_{A \rightarrow B} &= m\Delta v \end{aligned}$$

* Relationship b/w Gravitational field & Gravitational

Potential :- ($\vec{E}$ & V) :-

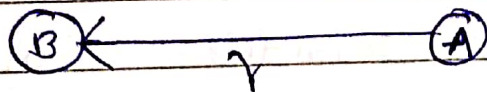
$$\sum \vec{E} = -\frac{dV}{dr}$$

In particular direction,

$$\sum E_x = -\frac{\partial V}{\partial x}, \quad E_y = -\frac{\partial V}{\partial y}, \quad E_z = -\frac{\partial V}{\partial z}$$

Derivation-

$$\begin{array}{c} \xrightarrow{\quad\quad\quad} E \\ -mE \xleftarrow{F_{ext}} (m) \xrightarrow{F_g} = mE \end{array}$$



$$W_{A \rightarrow B} = m \Delta E \text{ (fix disp)}$$

$$W_{A \rightarrow B} = -m \vec{E} \cdot \vec{r}$$

$$\therefore V_B - V_A = \frac{W_{A \rightarrow B}}{m} = \frac{-m E r}{m}$$

$$V_B - V_A = -E r$$

If, Distance b/w A & B is small, in potential Difference will also be small,

$$dV = -E dr$$

$$\therefore \sum \vec{E} = -\frac{dV}{dr}$$

*.) Gravitational Potential And Field :-

1.) For Hollow Sphere :-

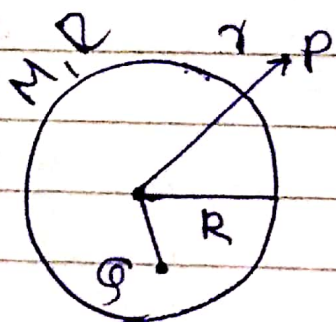
Or Shell :-

Hollow Sphere

Outside
($r > R$)

Surface
($R = r$)

Inside
($r < R$)



$\vec{E} :-$

$$\frac{-GM}{r^2}$$

$$\frac{-GM}{R^2}$$

0

$V :-$

$$\frac{-GM}{r}$$

$$\frac{-GM}{R}$$

$$\frac{-GM}{R}$$

2.) For Solid Sphere :-

($r > R$)

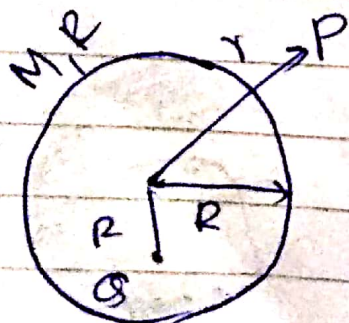
Outside

($r = R$)

Surface

($r < R$)

Inside



$\vec{E} :-$

$$\frac{-GM}{r^2}$$

$$\frac{-GM}{R^2}$$

$$\frac{-GM r}{R^3}$$

$V :-$

$$\frac{-GM}{r}$$

$$\frac{-GM}{R}$$

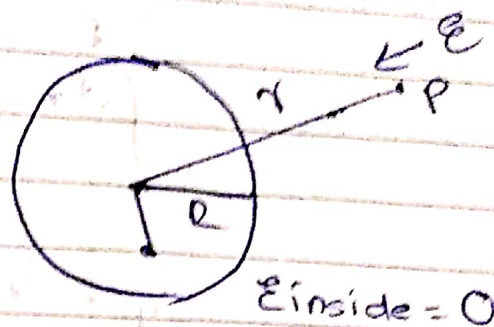
$$\frac{-GM}{R^3} \left[\frac{3R^2 - r^2}{2} \right]$$

★ Derivations —

1. For Hollow Sphere —

• Gravitational field at P: $-\frac{GM}{r^2}$

and, $E = -\frac{dv}{dr}$



For gravitational Potential,

$$\int_{\infty}^P dv = \int_{\infty}^P -E dr$$

$$= V_P - V_{\infty} = \int_{\infty}^r \frac{GM}{r^2} dr$$

$$V_P - V_{\infty} = GM \left[-\frac{1}{r} \right]_{\infty}^r = GM \left[-\frac{1}{r} - \left(-\frac{1}{\infty} \right) \right]$$

$$\therefore \left\{ V_P = -\frac{GM}{r} \right\} \dots \dots \text{(for outside)}$$

• at Surface, $r = R$

∴ Gravitational Potential at Surface

$$= \left\{ V_S = -\frac{GM}{R} \right\} \dots \dots \text{(at Surface)}$$

Inside sphere,
 $\vec{E} = 0$.

$$\therefore \int_s^p dr = \int_s^p -E dr = \int_s^p 0 dr = 0.$$

$$\therefore V_p - V_s = 0$$

$$\therefore V_p = V_s$$

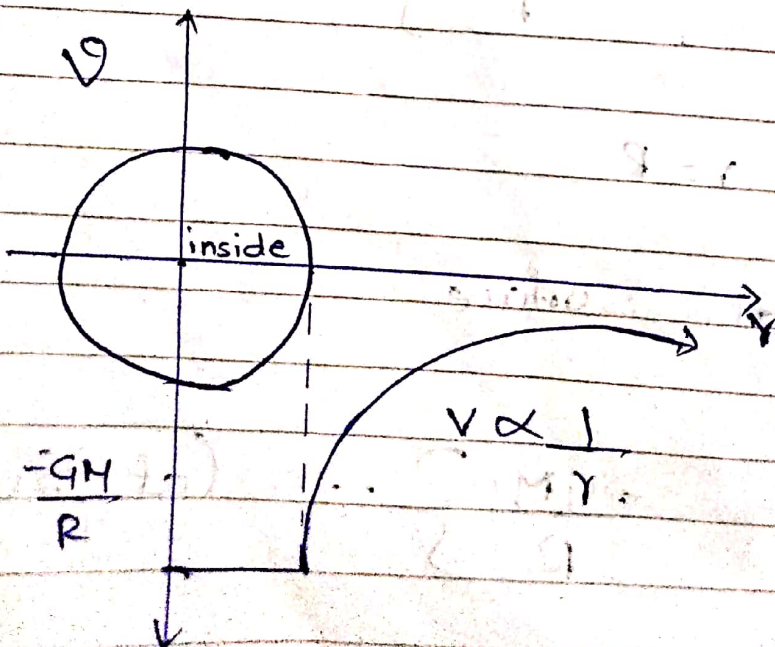
$$\therefore \left\{ V_i = V_p = V_s = -\frac{GM}{R} \right\}$$

$$\therefore E = -\frac{dv}{dr}, \quad 0 = -\frac{dv}{dr}$$

$$dv = 0 \quad \therefore v = \text{constant}$$

$\therefore$ Potential from Surface of Earth to inside and at center the Gravitational potential remains constant i.e. $-GM/R$.

Graph:-



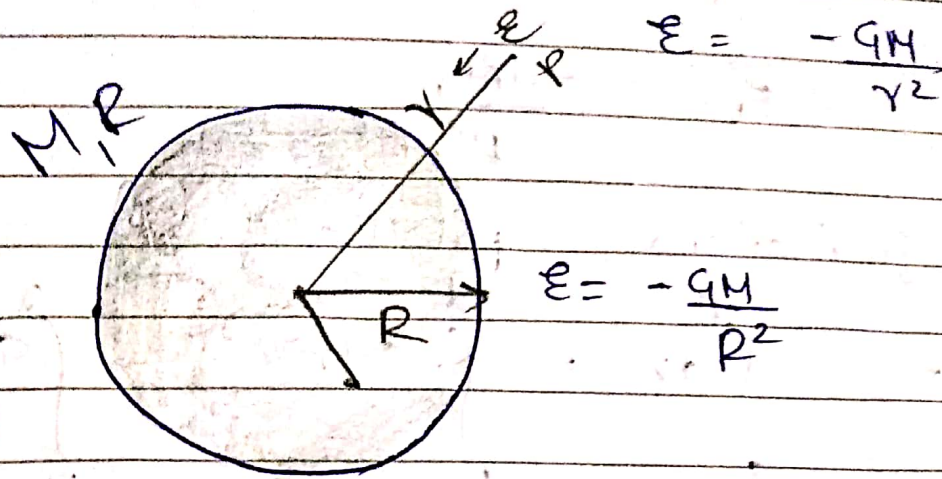
$$\bullet V_o = -\frac{GM}{R}$$

$$\therefore V_o \propto \frac{1}{r}$$

$$\bullet V_s = -\frac{GM}{R}$$

$$\bullet V_i = -\frac{GM}{R}$$

for Solid Sphere :-



• Outside

$$E = -\frac{GM}{r^2}$$

potential = $\int_{\infty}^r dV = \int_{\infty}^r -E dr$

$$\therefore V_p - V_{\infty} = \int_{\infty}^r \frac{GM}{r^2} dr$$

$$\therefore V_p - V_{\infty} = -\frac{GM}{r}$$

$$\therefore V_p = -\frac{GM}{r}$$

• At Surface

the r will be equal to R

$$\text{potential} = \therefore V_s = -\frac{GM}{R}$$

• Inside Sphere,

$$\vec{E}_i = -\frac{GM r}{R^3}$$

Gravitational potential,

$$= \int_S^P dv = \int_S^P -E dr = \int_R^r \frac{GM r dr}{R^3}$$

$$= V_P - V_S = \frac{GM}{R^3} \left[\frac{r^2}{2} \right]_R^r$$

$$\therefore V_P - V_S = \frac{GM}{R^3} \left[\frac{r^2}{2} - \frac{R^2}{2} \right]$$

$$\therefore V_S = -\frac{GM}{R}$$

$$\therefore V_P - \left(-\frac{GM}{R} \right) = \frac{GM}{R^3} \left[\frac{r^2 - R^2}{2} \right]$$

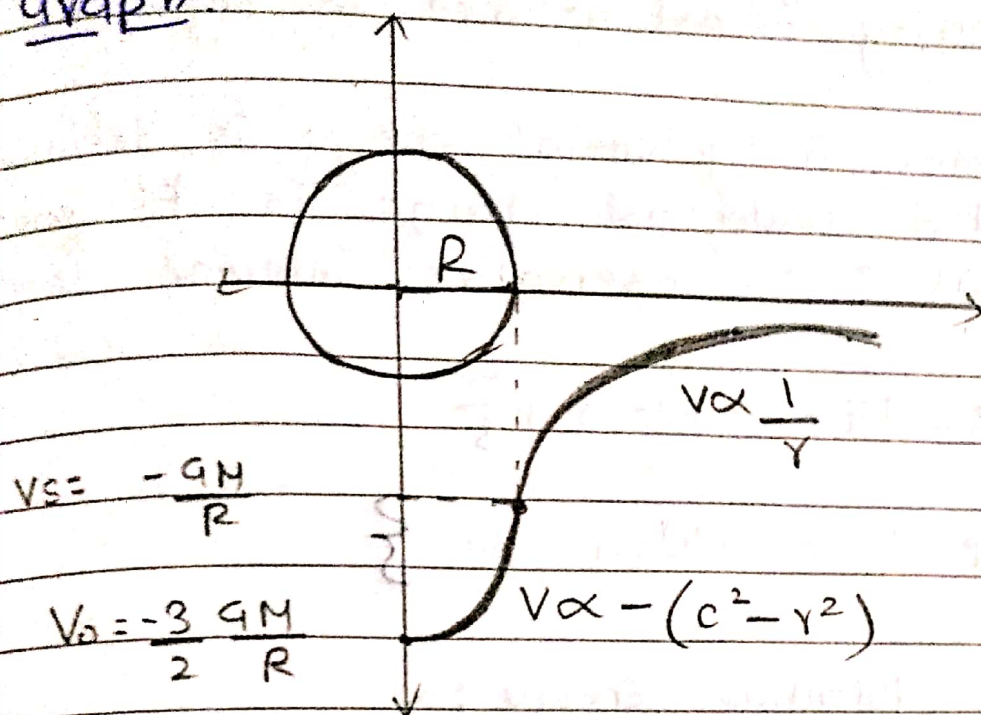
$$\therefore V_P = \frac{GM}{R^3} \left[\frac{r^2 - R^2}{2} \right] - \frac{GM}{R}$$

taking $\frac{GM}{R^3}$ common & multiplying by R^2 ,

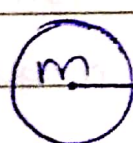
$$\therefore V_P = -\frac{GM}{R^3} \left[\frac{r^2 - R^2}{2} \right] + R^2$$

$$\left\{ \therefore V_P = \frac{GM}{R^3} \left[\frac{3R^2}{2} - \frac{1}{2} r^2 \right] \right\}$$

Graph:-



k) Gravitational Potential Energy :- ($\vec{E}$) :-

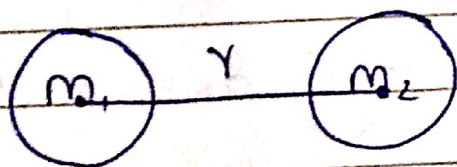


$$\vec{E} = -\frac{Gm}{r^2}$$

$$V = -\frac{GM}{r}$$

$$U = 0$$

Gravitational Potential Energy for a single mass is 0



here,

Gravitational Potential Energy

$$\left\{ = -\frac{Gm_1 m_2}{r} \right\}$$

Potential Energy is not defined in Physics.

But difference in potential energy is defined.
provided k.E should not change is $k.E = \text{constant}$
i.e. $\sum \Delta U = -W_{\text{conservative internal force}}$

$$\text{i.e. } \{ U_f - U_i = -W_{c. f.} \}$$

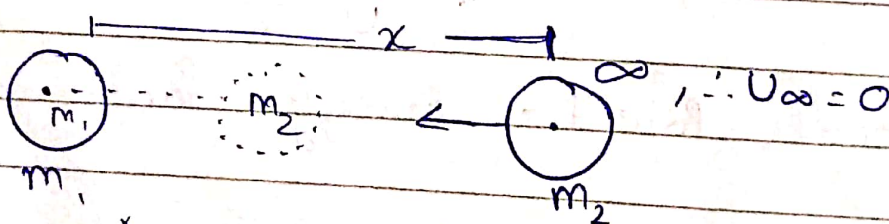
$$= \{ U_f - U_i = -W_{\text{grav. force}} \}$$

Gravitational Potential Energy :-

At ∞ ; $U_{\infty} = 0$. (Assumed)

$$= \{ U_{\text{config}} - U_{\infty}^0 = -W_{\text{grav. force in bringing each mass from } \infty} \}$$

★) Derivation :-

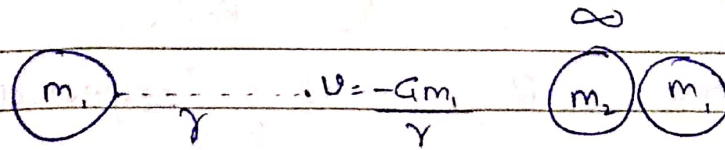


$$\begin{aligned} \therefore W &= \int_{\infty}^r f \cdot dx = \int_{\infty}^r -\frac{Gm_1 m_2}{x^2} dx = -Gm_1 m_2 \int_{\infty}^r \frac{dx}{x^2} \\ &= Gm_1 m_2 \left[\frac{1}{x} \right]_{\infty}^r = \frac{Gm_1 m_2}{r} \end{aligned}$$

$$\therefore \{ U_{\text{config}} = -\frac{Gm_1 m_2}{r} \}$$

$$\{\Delta U = +W_{ext}\}$$

Derivation without integration:-



$$\begin{aligned} W_2 &= m \Delta U \\ &= m_2 (V_r - V_{\infty}) \\ &= m_2 \left(-\frac{Gm_1}{r} - 0 \right) \end{aligned}$$

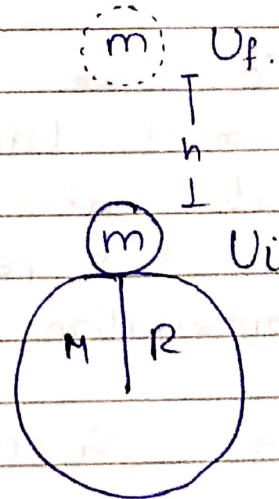
$$\left\{ W_2 = -\frac{Gm_1m_2}{r} \right\}$$

*) Potential Energy :-

We know $g = \frac{GM}{R^2}$

$$U_i = -\frac{GmM}{R}$$

$$U_f = -\frac{GmM}{(R+h)}$$



$$\therefore \Delta U = -\frac{GmM}{R} - \left(-\frac{GmM}{(R+h)} \right)$$

$$= -\frac{GmM}{R} + \frac{GmM}{R+h}$$

$$= -GmM \left(\frac{-1}{R} + \frac{1}{R+h} \right)$$

$$= -GmM \left(\frac{-R + R+h}{R(R+h)} \right) = -\frac{GmMh}{R(R+h)}$$

$$\therefore GM = gR^2$$

$$\left\{ \therefore \frac{gR^2mh}{R(R+h)} = \frac{mgh}{\left(1 + \frac{h}{R}\right)} \right\}$$

$\therefore h \ll R$, it is neglected $\frac{h}{R} \rightarrow 0$

$$\therefore \left\{ \Delta U = mgh \right\}_{U_H - U_S}$$

6.) Escape Velocity :-

Earth's gravitational field end at ∞

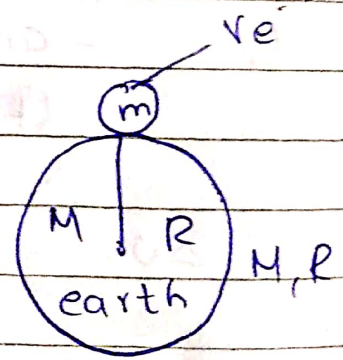
To give a body its escape velocity, we must throw it with a speed such that it just escapes earth's gravitational field.
To escape $\rightarrow$ To ∞ (Total energy = 0)

•) Conservation of mechanical Energy :-

i.e $U_i + K_i = U_f + K_f$

$$= -\frac{GMm}{R} + \frac{1}{2}mv_e^2 = 0 + 0$$

$$= \therefore \frac{1}{2}mv_e^2 = \frac{GMm}{R}$$



$$\left\{ \therefore v_e = \sqrt{\frac{2GM}{R}} \right\}$$

Valid only when body is escaping from surface.

• Valid for any planet

Escape Velocity for Earth:-

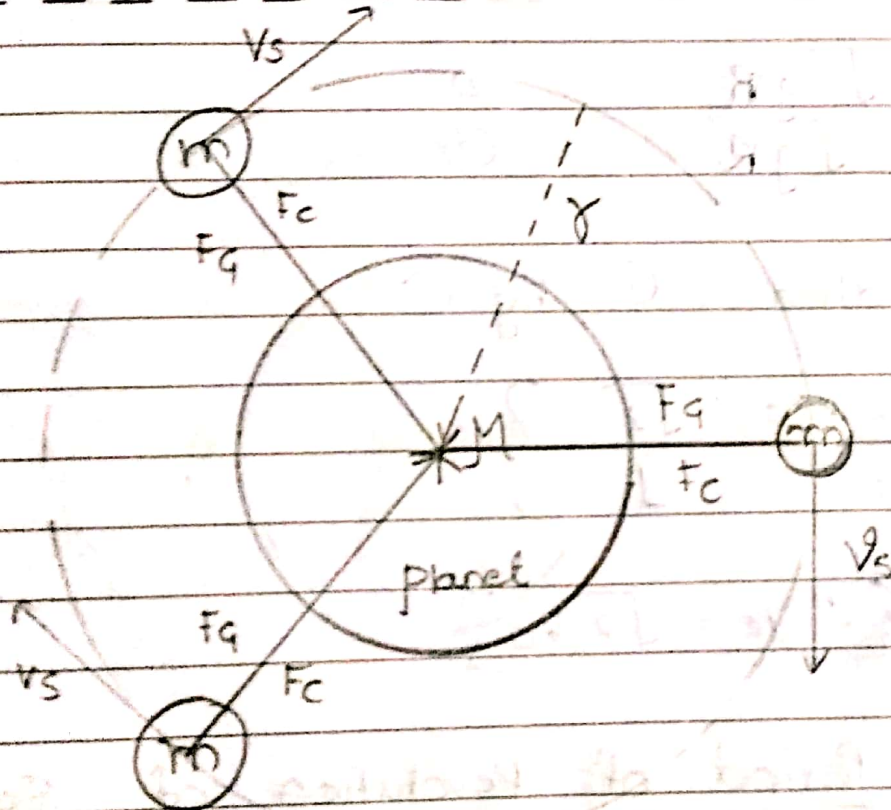
$$v_e = \sqrt{\frac{2GM}{R}}$$

$$v_e = \sqrt{\frac{2gR^2}{R}}$$

$$\{ v_e = \sqrt{2gR} \} = \{ 11.2 \text{ km/s} \}$$

*) Motion of Satellites :-

1.) Speed of Satellites :-



$$F_c = \frac{mv^2}{r}$$

$$F_g = \frac{GMm}{r^2}$$

$$\frac{mv^2}{r} = \frac{GMm}{r^2}$$

$$\{ v = \sqrt{\frac{GM}{r}} \} \rightarrow \text{Orbital Speed}$$

If Satellite is revolving very close to surface

i.e $r = R$

$$\therefore V_o = \sqrt{\frac{GM}{R}} = \sqrt{\frac{gR^2}{R}}$$

$$\therefore \{ V_o = \sqrt{gR} \} \rightarrow \text{Orbital Speed of Earth}$$

$$\{ = 7.9 \text{ km/sec or } 8 \text{ km/sec} \}$$

*.) Relationship blw V_o And V_e :-

$$V_e = \sqrt{2gR} \quad \dots \textcircled{1}$$

$$V_o = \sqrt{gR} \quad \dots \textcircled{2}$$

Dividing $\textcircled{1}$ by $\textcircled{2}$,

$$\left\{ \frac{V_e}{V_o} = \frac{\sqrt{2}}{1} \right\}$$

$$\{ \therefore V_e = \sqrt{2} V_o \}$$

*) Time Period of Revolution of Satellite :-

$$\text{Time period} = \frac{\text{Distance}}{\text{Speed}} = \frac{2\pi r}{V}$$

$$\left\{ = \frac{2\pi r^1}{\sqrt{\frac{GM}{r}}} = T = \frac{2\pi r^{1+\frac{1}{2}}}{\sqrt{GM}} = \frac{2\pi r^{\frac{3}{2}}}{\sqrt{GM}} \right\}$$

$$\left\{ T = \frac{2\pi r^{\frac{3}{2}}}{\sqrt{gR^2}} = T = K r^{\frac{3}{2}} \right\}$$

$$\left\{ \text{i.e. } T \propto r^{\frac{3}{2}} = T^2 \propto r^3 \right\}$$

* Time period of Revolution of Satellite close to Earth:-

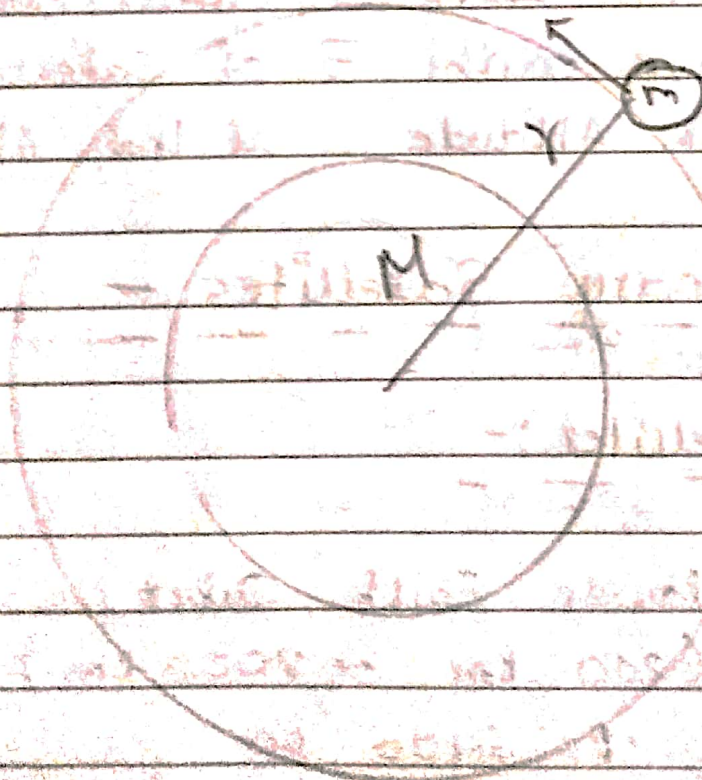
$r \rightarrow R$

$$\therefore T = \frac{2\pi r}{v} = \frac{2\pi R}{\sqrt{\frac{GM}{R}}}$$

$$\left\{ T = \frac{2\pi R}{\sqrt{\frac{gR}{R}}} = \frac{2\pi R}{\sqrt{g}} = 2\pi \sqrt{\frac{R}{g}} \right\}$$

comes out to be 84.6 min.

* Energy of a Satellite :-



If $TE = -ve$ (Bounded system) (Circle/ellipse)

If $TE = 0$, (free system) (Parabola)

If $TE = +ve$ (free system) (Hyperbola)

Page No.	
Date	

$$K.E = \frac{1}{2}mv^2$$

$$= \frac{1}{2}m \left(\sqrt{\frac{GM}{r}} \right)^2 = \frac{1}{2} \frac{mGM}{r}$$

$$\left\{ \therefore K.E = \frac{GMm}{2r} \right\} \quad \text{--- (1)}$$

and potential Energy,

$$\left\{ U = -\frac{GMm}{r} \right\} \quad \text{--- (2)}$$

$$\therefore \text{Total energy} = (1) + (2) = K.E + U$$

$$\left\{ \therefore TE = -\frac{GMm}{2r} \right\} \quad \left\{ \text{Binding Energy} \right\}$$

Total Energy is negative becaz satellite revolves in a bounded Curve.

$$\left\{ \begin{array}{l} \text{Total Energy of Star} \\ \text{Satellite above earth} \\ \text{Surface at An Altitude} \end{array} \right\} = \left\{ \begin{array}{l} \text{Total Energy of Satellite} \\ \text{at that Altitude} \end{array} \right\} - \left\{ \begin{array}{l} \text{Total Energy} \\ \text{at Surface} \end{array} \right\}$$

* Geo-Stationary Satellites ---

1.) Types of Satellites :-

a.) L.E.O --- (Lower Earth Orbit)

• 1 day = 12 rev. (200 km ~ 2000 km)

• $R = 6400$ km

• $T = 1.5 \sim 2$ hrs

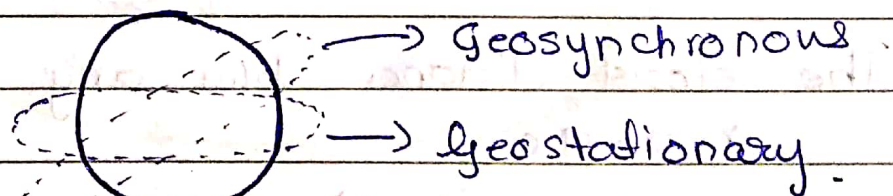


- b.) M.E.O - (Middle Earth Orbit)
- (20,000 km) , • (2000 km ~ 36,000 km)
 - $T = 12$ hr
 - 1 day = 2 revolutions

- c.) G.E.O - (Geostationary orbit)
- (36,000 km)
 - $T = 24$ hr
 - 1 day = 1 revolution

*.) Geostationary Satellites :-

- 1 revolution in 24 hrs.
- Which revolves around Earth such that it appears stationary from any point of Earth.
- Which has same sense of rotation as that of Earth.
- Which is placed at an altitude of 36,000 km above earth.
- Geostationary Satellites revolve around equator belt.
- 'Geosynchronous Satellite' ($T=24$ hr, $h=36,000$ km), but inclined.



• Velocity of G.E.S - $F_a = F_g$

$$= \frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$\therefore v = \sqrt{\frac{GM}{r}}$$

• Time period of satellites

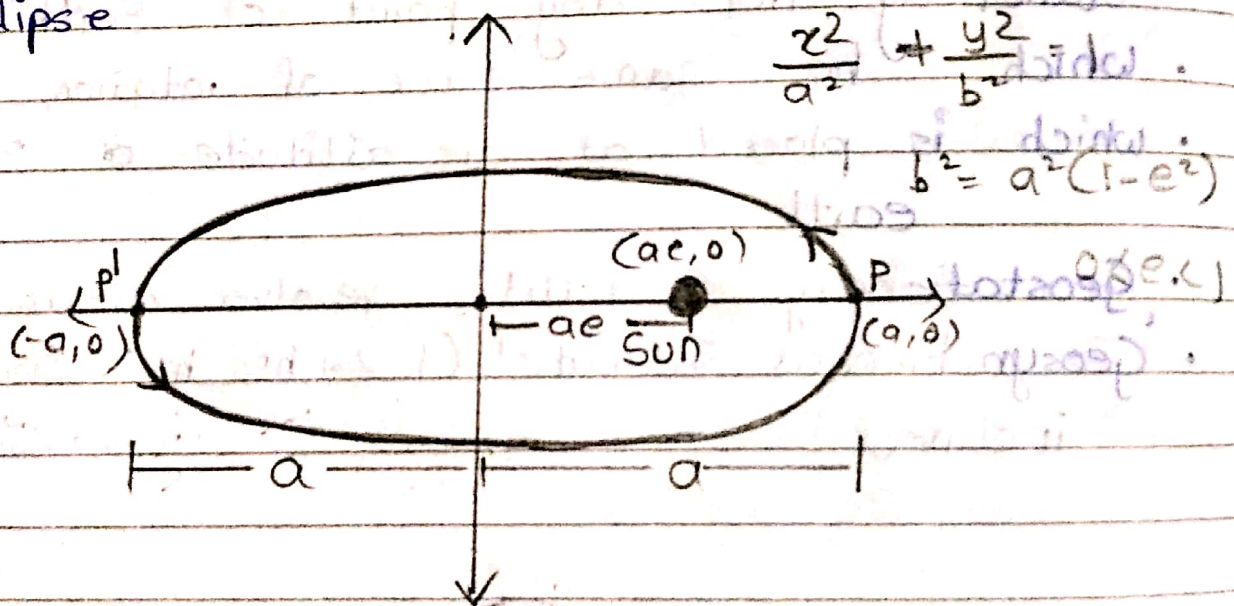
$$= \frac{2\pi r}{v} = \frac{2\pi r}{\sqrt{\frac{GM}{r}}}$$

$$\left\{ T = \frac{2\pi r^{1+\frac{1}{2}}}{\sqrt{GM}} = \frac{2\pi r^{3/2}}{\sqrt{gR^2}} \right\}$$

$$\left\{ \begin{array}{l} \text{i.e. } T = kr^{3/2} \\ T^2 \propto r^3 \end{array} \right\}$$

*.) Kepler's law of planetary Motion :-

1.) First law - All planets revolve around sun in closed elliptical orbits & sun is at one focus of ellipse



The closest distance b/w any planet & sun is,
 $r = a - ae$

or $r = a(1-e)$ called as 'Perihelion or Perigee'

The farthest distance b/w sun & planet is,
 $r = a + ae$

$r = a(1+e)$ called as 'Aphelion' or 'Aphogee'

2.) Second law - Planets do not revolve around sun with constant speeds.

i.e Planets move faster when it is near to sun and slower when it is farthest from sun.

— 'Actual law' :—

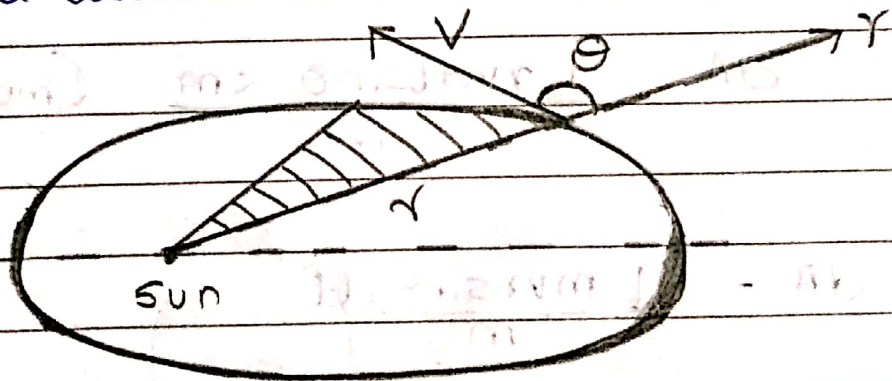
'The Areal Velocity of a planet is constant.'

or
'A planet sweeps equal area in equal time around Sun.'

i.e $\frac{dA}{dt} = \text{constant}$

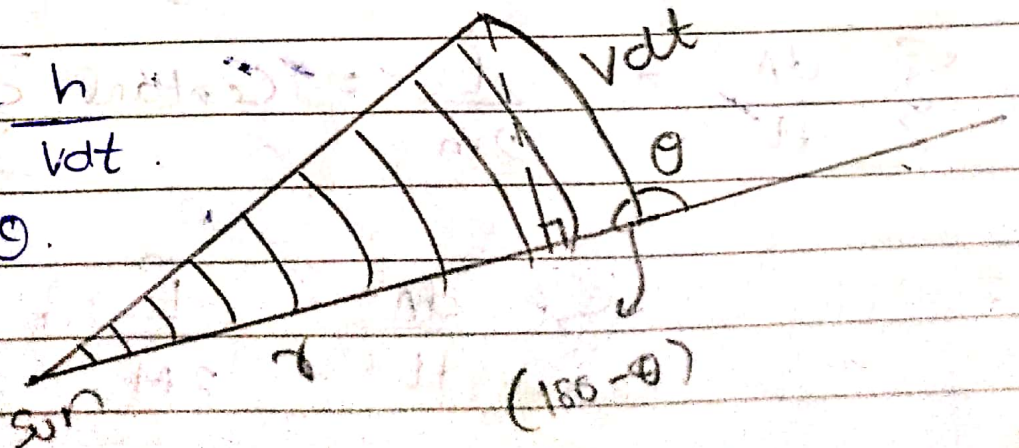
or $\frac{A_1}{t_1} = \frac{A_2}{t_2} = \frac{A_3}{t_3} = \text{constant}$

*) Proof for Kepler's Second law :—



$$\sin(180 - \theta) = \frac{h}{v dt}$$

$$\therefore h = v dt \sin \theta$$



Planet revolving around Sun has
constant Angular momentum.

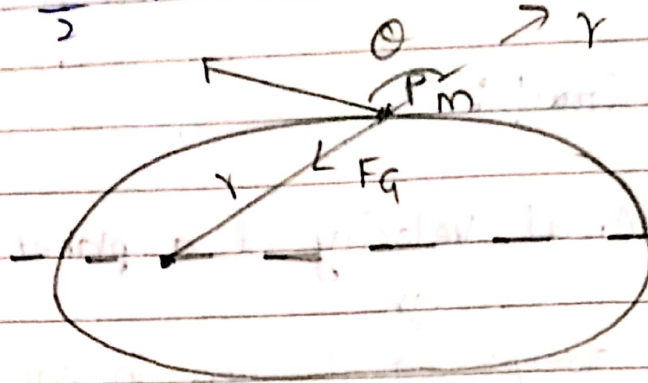
Page No.			
Date			

Planet moves for time 'dt'

∴ Distance covered by Planet = $V dt$

$$\text{Area} = \frac{1}{2} \times b \times h$$

$$dA = \frac{1}{2} \times r \times V dt \sin \theta \quad \dots \quad (1)$$



$$\tau = \vec{r} \times \vec{F} = 0$$

$$= r F \sin 0 = 0.$$

if $\tau_{\text{net}} = 0$
then $\vec{L} = \text{Constant}$

i.e. $\vec{L} = \vec{r} \times \vec{p}$

$$= \vec{r} \times m \vec{v}$$

$$L = m v r \sin \theta = \text{Constant} \quad \dots \quad (2)$$

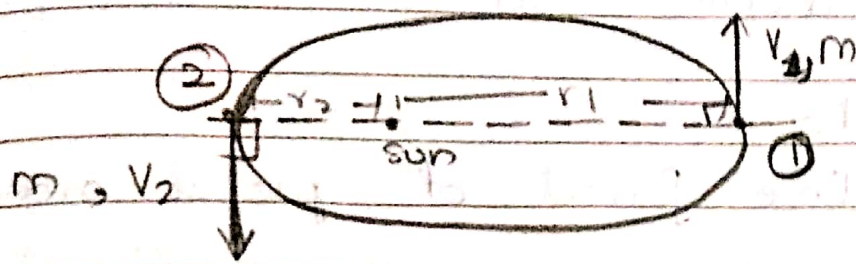
$$\therefore dA = \frac{1}{2} \frac{r v dt \sin \theta \times m}{m} \quad (\text{multiplying eqn (1) by } \frac{m}{m})$$

$$dA = \frac{1}{2} \frac{m v r \sin \theta dt}{m}$$

$$\left\{ \frac{dA}{dt} = \frac{\vec{L}}{2m} = \text{Constant} \right\}$$

$$\therefore \left\{ \frac{dA}{dt} = \frac{\vec{L}}{2M} = \text{Constant} \right\}$$

* Speed of Planets at aphogee & Perigee:-



∴ Angular momentum is 0, i.e. $\vec{L} = 0$, becaz $\tau = 0$, (about sun),

then, $L_1 = L_2$

$$r_1 \times p_1 = r_2 \times p_2$$

$$r_1 \times m\vec{v}_1 = r_2 \times m\vec{v}_2$$

$$v_1 r_1 = v_2 r_2$$

$$r_1 = a(1+e)$$

$$r_2 = a(1-e)$$

$$\therefore v_1 a(1+e) = v_2 a(1-e) \dots \dots \textcircled{1}$$

Applying Conservation of energy at pt ① & ②,

$$\text{i.e. } T.E_1 = T.E_2$$

$$= U_1 + K_1 = U_2 + K_2$$

$$= \frac{-GM_s m}{r_1 a(1+e)} + \frac{1}{2} m v_1^2 = \frac{-GM_s m}{a(1-e)} + \frac{1}{2} m v_2^2 \dots \dots \textcircled{2}$$

on solving eqⁿ ① & ②,

$$v_1 = \sqrt{\frac{GM_s}{a} \left(\frac{1-e}{1+e} \right)} \quad \& \quad v_2 = \sqrt{\frac{GM_s}{a} \left(\frac{1+e}{1-e} \right)}$$

(Aphogee) (Perigee)

Third law :-

let us assume planet moves in a circular orbit around Sun

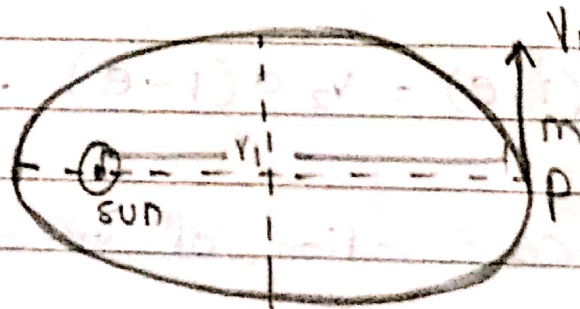
$$\therefore F_c = F_g$$

Then time period of planet around Sun

$$\left\{ \begin{array}{l} T = \frac{2\pi r^{\frac{3}{2}}}{\sqrt{GM_s}} \\ T \propto r^{\frac{3}{2}} \end{array} \right\}$$

Actual Proof :-

$$(T^2 \propto a^3)$$



We know , $\frac{dA}{dt} = \frac{L}{2m}$

1 complete revolution
= Area Swept = Area of ellipse = πab
time = T

$$\therefore \frac{\pi ab}{T} = \frac{L}{2m}$$

$$= \frac{\pi ab}{T} = \frac{r_1 m v_1}{2m}$$

①

$$\text{Angular momentum} = \vec{r} \times \vec{p} = r_1 m v_1$$

$$\therefore r_1 = a + ae = a(1+e)$$

$$\therefore v_1 = \sqrt{\frac{GM_s}{a} \left(\frac{1-e}{1+e} \right)}$$

putting this value is eqⁿ ①, $\therefore$ Angular momentum is constant.

$$\therefore \frac{\pi a b}{T} = a(1+e) \sqrt{\frac{GM_s}{a} \left(\frac{1-e}{1+e} \right)}$$

Squaring both sides,

$$\frac{\pi^2 b^2}{T^2} = \frac{(1+e)^2 GM_s (1-e)}{a (1+e)} \quad [\because b^2 = a^2(1-e^2)]$$

$$= \frac{\pi^2 a^3 (1-e^2)}{T^2} = (1+e) GM_s (1-e)$$

$$\therefore T^2 = \frac{\pi^2 a^3}{GM_s}$$

$$\left\{ \therefore T^2 \propto a^3 \right\}$$

Hence proved